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ALPHA PLUS
(PUNJABI BAGH)
9310030102

ANALYSIS - II

LIMITS AND CONTINUITY

NEW SYLLABUS (2011-2012)

SECTION - 1 (Chapter - 4)

SEMESTER - II MATHS (H)

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NOTE : * * * * *
NEW PROBABILITY & STATISTICS
NOTES ALSO AVAILABLE AT
ALPHA PLUS CENTERS

Reference book :

R. G. BARTLE

D. R. SHERBERT

(THIRD EDITION)

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Opp. McDonalds

Tel. 011 42467484, 9899991900

Kupang P.D.
Wed And
Fri 8:00
SAT 8:00
Sun 8:00

LIMITS Ch. 4

21

Def: CLUSTER POINT

Let $A \subseteq \mathbb{R}$. A point $c \in \mathbb{R}$ is a cluster point of A if for every $\delta > 0$, there exist at least one point $x \in A$, $x \neq c$ s.t. $|x - c| < \delta$

Def: A point $c \in \mathbb{R}$ is a cluster point of the set A if $\forall \delta > 0$, there exist $]c - \delta, c + \delta[$ contains at least one point of $A \setminus \{c\}$

Thm-1 A number $c \in \mathbb{R}$ is a cluster point of a subset A of $\mathbb{R}$ iff there exist a seq $\langle a_n \rangle$ in A s.t. $\lim_{n \rightarrow \infty} a_n = c$ and $a_n \neq c \quad \forall n \in \mathbb{N}$

Proof: [To show: $\exists$ sequence $\langle a_n \rangle$ in A with $a_n \neq c \quad \forall n \in \mathbb{N}$ as $\lim a_n = c$]

As c is a limit point A $\therefore \forall \frac{1}{n} > 0, \exists c \neq a_n$
 $a_n \in A$ s.t. $|a_n - c| < \frac{1}{n} \quad \forall n \in \mathbb{N}$

$$\Rightarrow \lim a_n = c$$

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Conversely: let us assume that $\exists$ a sequence $\langle a_n \rangle$ in A with $a_n \neq c \quad \forall n \in \mathbb{N}$ as $\lim a_n = c$

[To show: c is a cluster point of A]

Let $\epsilon > 0$ be any number, as $\langle a_n \rangle \rightarrow c$

$\therefore$ by definition $\exists k(\epsilon) \in \mathbb{N}$, $|a_n - c| < \epsilon \quad \forall n \geq k(\epsilon)$

$$\Rightarrow a_n \in]c - \epsilon, c + \epsilon[\quad \forall n \geq k(\epsilon) \text{ with } a_n \neq c \quad \forall n \in \mathbb{N}$$

$\Rightarrow V_\epsilon(c) =]c - \epsilon, c + \epsilon[$ contains infinitely many points of A distinct from c

$\Rightarrow]c - \epsilon, c + \epsilon[$ contains at least one point of A distinct from c

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P.2

- 1) examples : A finite set has no cluster points.
- 2) The infinite set $\mathbb{N}$ has no cluster point.

LIMIT OF A FUNCTION :

Defn : Let $A \subseteq \mathbb{R}$ & c be a cluster pt. of A

Let $f: A \rightarrow \mathbb{R}$ & let $l \in \mathbb{R}$

Then $\lim_{x \rightarrow c} f(x) = l$ iff for each $\epsilon > 0$, $\exists \delta > 0$
s.t. $x \in A$, $0 < |x - c| < \delta \Rightarrow |f(x) - l| < \epsilon$

$\forall x \in V_\delta(c)$

Thm-2 If $f: A \rightarrow \mathbb{R}$ and if c is a cluster point of A , then f can have only one limit at c

Proof: Let $\lim_{x \rightarrow c} f(x) = l$ & $\lim_{x \rightarrow c} f(x) = l'$

[To show $l = l'$]

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Let $\epsilon > 0$ be given.

$\lim_{x \rightarrow c} f(x) = l$ iff for each $\frac{\epsilon}{2} > 0$, $\exists \delta_1 > 0$ s.t.

$$|x - c| < \delta_1 \Rightarrow |f(x) - l| < \frac{\epsilon}{2} \quad \text{--- (1)}$$

$\lim_{x \rightarrow c} f(x) = l'$ iff for each $\frac{\epsilon}{2} > 0$, $\exists \delta_2 > 0$ s.t.

$$|x - c| < \delta_2 \Rightarrow |f(x) - l'| < \frac{\epsilon}{2} \quad \text{--- (2)}$$

let $\delta = \min(\delta_1, \delta_2)$.

$$\left. \begin{array}{l} |f(x) - l| < \frac{\epsilon}{2} \\ |f(x) - l'| < \frac{\epsilon}{2} \end{array} \right\} \forall x \in]c - \delta, c + \delta[\quad \text{--- (3)}$$

$$\begin{aligned} \text{Consider } |l - l'| &= |l - f(x) + f(x) - l'| \\ &\leq |f(x) - l| + |f(x) - l'| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \quad \text{from (1)} \\ &< \epsilon \end{aligned}$$

$$\Rightarrow l = l'$$

SEQUENTIAL CRITERION FOR LIMITS :

Theorem 3: Let $A \subseteq \mathbb{R}$ & c be a cluster point of A

Let $f: A \rightarrow \mathbb{R}$ & let $l \in \mathbb{R}$

Then the following are equivalent

(i) $\lim_{x \rightarrow c} f(x) = l$

(ii) for every sequence $\langle x_n \rangle$ in A s.t. $\langle x_n \rangle \rightarrow c$
(with $x_n \neq c \forall n \in \mathbb{N}$) the sequence $\langle f(x_n) \rangle \rightarrow l$

Proof (i) $\Rightarrow$ (ii)

Given : $\lim_{x \rightarrow c} f(x) = l$

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Let $\langle x_n \rangle$ be a sequence in A s.t. $\langle x_n \rangle \rightarrow c$
($x_n \neq c \forall n \in \mathbb{N}$)

To show $\langle f(x_n) \rangle \rightarrow l$. Let $\epsilon > 0$ be given

[T.s $\exists n_0 \in \mathbb{N}$ s.t. $|f(x_n) - l| < \epsilon \forall n \geq n_0$]

As $\lim_{x \rightarrow c} f(x) = l$ $\therefore$ corr to $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$x \in A, 0 < |x - c| < \delta \Rightarrow |f(x) - l| < \epsilon \quad \text{--- (1)}$$

As $\langle x_n \rangle \rightarrow c$ $\therefore$ corr to $\delta > 0$, $\exists n_0 \in \mathbb{N}$ s.t.

$$|x_n - c| < \delta \quad \forall n \geq n_0$$

Also, $x_n \neq c$ i.e. $|x_n - c| > 0 \quad \forall n \in \mathbb{N}$

Thus $x_n \in A$, $0 < |x_n - c| < \delta \quad \forall n \geq n_0$

$$\therefore |f(x_n) - l| < \epsilon \quad \forall n \geq n_0 \quad \text{by (1)}$$

$$\therefore \langle f(x_n) \rangle \rightarrow l$$

(ii) $\Rightarrow$ (i)

Let condition (ii) be given. To show $\lim_{x \rightarrow c} f(x) = l$

i.e. To show : for each $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$0 < |x - c| < \delta \Rightarrow |f(x) - l| < \epsilon$$

AL Possible. Suppose this is false.

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P.4

In particular, taking $\delta = \frac{1}{n}$ (for each $n \in \mathbb{N}$), we get:

$\exists x_n \in A$ s.t. for each $\delta > 0$, $0 < |x_n - c| < \frac{1}{n}$ but $|f(x_n) - l| \geq \epsilon$

Now $|x_n - c| < \frac{1}{n} \Rightarrow \langle x_n \rangle \rightarrow c$

• $|x_n - c| > 0 \Rightarrow x_n \neq c \quad \forall n \in \mathbb{N}$

$\therefore$ by condition (ii), $\langle f(x_n) \rangle \rightarrow l$

$\therefore$ Contr to above $\epsilon > 0$, $\exists n_0 \in \mathbb{N}$ s.t. $|f(x_n) - l| < \epsilon \quad \forall n \geq n_0$

Which contradicts ③

$\therefore \lim_{x \rightarrow c} f(x) = l$

PROBLEM SET - I

ex 1 Show $\lim_{x \rightarrow c} x^2 = c^2$

Soln: Let $f(x) = x^2$

To show $\lim_{x \rightarrow c} f(x) = c^2$

Let $\epsilon > 0$ be given

[To show $\exists \delta > 0$ s.t. $0 < |x - c| < \delta \Rightarrow |f(x) - c^2| < \epsilon$]

$$\begin{aligned} \text{consider } |f(x) - c^2| &= |x^2 - c^2| \\ &= |x - c||x + c| \quad \text{--- ①} \end{aligned}$$

Let us take $|x - c| < 1$

$$|x| = |x - c + c| \leq |x - c| + |c| < 1 + |c|$$

$$|x + c| \leq |x| + |c| < 1 + |c| + |c| = K, \text{ say, where } K > 0$$

$$\therefore \text{① gives: } |f(x) - c^2| < K|x - c| \quad \text{--- ②}$$

Let $\delta = \min \left\{ 1, \frac{\epsilon}{K} \right\}$

consider $0 < |x - c| < \delta$:

$$\therefore |x - c| < 1 \text{ and } |x - c| < \frac{\epsilon}{K}$$

$$\therefore |x - c| < \frac{\epsilon}{K} \Rightarrow K|x - c| < \epsilon$$

$$\therefore |f(x) - c^2| < \epsilon$$

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ex 2) Show $\lim_{x \rightarrow c} \frac{1}{x} = \frac{1}{c}$ for $c > 0$

soln: Let $f(x) = \frac{1}{x}$

To show $\lim_{x \rightarrow c} f(x) = \frac{1}{c}$

Let $\epsilon > 0$ be given

[To show $\exists \delta > 0$ s.t. $x \in \mathbb{R}^+$, $0 < |x - c| < \delta \Rightarrow |f(x) - \frac{1}{c}| < \epsilon$]

$$\text{consider } |f(x) - \frac{1}{c}| = \left| \frac{1}{x} - \frac{1}{c} \right| = \left| \frac{c - x}{xc} \right| = \frac{1}{c} \cdot \frac{1}{x} |x - c| \quad \text{--- (1)}$$

Let us take $|x - c| < \frac{\epsilon}{2}$

$$\text{consider } c - x \leq |c - x| < \frac{\epsilon}{2} \quad \therefore x > c - \frac{\epsilon}{2}$$

$$1 < x > \frac{c}{2} \quad \therefore \frac{1}{x} < \frac{2}{c}$$

$$\therefore \text{(1) gives } |f(x) - \frac{1}{c}| < \frac{1}{c} \cdot \frac{2}{c} |x - c| \quad \text{--- (2)}$$

$$\text{Let } \delta = \min \left\{ \frac{\epsilon}{2}, \epsilon \frac{c^2}{2} \right\}$$

Consider $x \in \mathbb{R}^+$, $0 < |x - c| < \delta$

$$\therefore |x - c| < \frac{\epsilon}{2} \quad \text{and} \quad |x - c| < \epsilon \frac{c^2}{2} \quad S$$

$$\therefore \text{(2) gives : } |f(x) - \frac{1}{c}| < \frac{2}{c} \cdot \left(\frac{\epsilon c^2}{2} \right) = \epsilon$$

ex 3) Show $\lim_{x \rightarrow c} \sqrt{x} = \sqrt{c}$ for $c > 0$.

soln: Let $f(x) = \sqrt{x}$

To show $\lim_{x \rightarrow c} f(x) = \sqrt{c}$

Let $\epsilon > 0$ be given

[To show $\exists \delta > 0$ s.t. $0 < |x - c| < \delta \Rightarrow |f(x) - \sqrt{c}| < \epsilon$]

$$\text{Consider } |f(x) - \sqrt{c}| = |\sqrt{x} - \sqrt{c}| = \frac{1}{\sqrt{x} + \sqrt{c}} |x - c| \quad \text{--- (1)}$$

$$\text{Now } \sqrt{x} + \sqrt{c} > \sqrt{c} \quad \therefore \frac{1}{\sqrt{x} + \sqrt{c}} < \frac{1}{\sqrt{c}}$$

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P.6

Let $\delta = \epsilon \sqrt{c}$

Consider $x \in \mathbb{R}^+$, $0 < |x - c| < \delta$ $\therefore |x - c| < \epsilon \sqrt{c}$

$\therefore$ ② gives : $|f(x) - \sqrt{c}| < \frac{1}{\sqrt{c}} \cdot \epsilon \sqrt{c} = \epsilon$

✓ ex 4). Show $\lim_{x \rightarrow 2} \frac{x^3 - 4}{x^2 + 1} = \frac{4}{5}$

Soln: Let $f(x) = \frac{x^3 - 4}{x^2 + 1}$

To show $\lim_{x \rightarrow 2} f(x) = \frac{4}{5}$

Let $\epsilon > 0$ be given

[To show $\exists \delta > 0$ s.t. $0 < |x - 2| < \delta \Rightarrow |f(x) - \frac{4}{5}| < \epsilon$]

consider $|f(x) - \frac{4}{5}| = \left| \frac{5x^3 - 4x^2 - 24}{5(x^2 + 1)} \right|$

Now $x = 2$ is a zero of $5x^3 - 4x^2 - 24$

$\therefore x - 2$ is a factor " " " "

By long-division : $5x^3 - 4x^2 - 24 = (x - 2)(5x^2 + 6x + 12)$

$\therefore |f(x) - \frac{4}{5}| = \frac{|5x^2 + 6x + 12|}{5(x^2 + 1)} |x - 2| \quad - (1)$

Let us take $|x - 2| < 1$ i.e. $2 - 1 < x < 2 + 1$ i.e. $1 < x < 3$

Now $x < 3 \Rightarrow 5x^2 + 6x + 12 < 75$

$x > 1 \Rightarrow 5(x^2 + 1) > 10$

$\therefore$ ① gives : $|f(x) - \frac{4}{5}| < \frac{75}{10} |x - 2| \quad - (2)$

Let $\delta = \min(1, \frac{2}{15} \epsilon)$

consider $0 < |x - 2| < \delta$:

$\therefore |x - 2| < 1$ & $|x - 2| < \frac{2}{15} \epsilon$

$\therefore$ ② gives : $|f(x) - \frac{4}{5}| < \frac{15}{2} \cdot \frac{2}{15} \epsilon = \epsilon$

As $\epsilon > 0$ be arbitrary $\therefore \lim_{x \rightarrow 2} f(x) = \frac{4}{5}$

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$$\text{Q9(4.1)(a)} \quad \lim_{x \rightarrow 2} \frac{1}{1-x} = -1$$

soln: Let $\epsilon > 0$ be given

$$\text{consider } \left| \frac{1}{1-x} + 1 \right| = \left| \frac{1+1-x}{1-x} \right| = \frac{|x-2|}{|x-1|} \quad \text{--- (I)}$$

since we are taking $\lim_{x \rightarrow 2} \therefore x > 1 \Rightarrow x-1 > 0$

$$\Rightarrow |x-1| = (x-1) \quad \text{--- (II)}$$

$$\text{Now if } |x-2| < \frac{1}{2} \Rightarrow \frac{3}{2} < x < \frac{5}{2} \Rightarrow \frac{3}{2}-1 < x-1 \\ \Rightarrow \frac{1}{2} < (x-1) \Rightarrow \frac{1}{x-1} < 2$$

$$\text{from (I) \& (II)} \quad \left| \frac{1}{1-x} + 1 \right| < 2|x-2|$$

$$\text{Now } \left| \frac{1}{1-x} + 1 \right| < \epsilon \text{ if } 2|x-2| < \epsilon \Rightarrow |x-2| < \frac{\epsilon}{2}$$

$$\text{Choose } \delta = \min \left\{ \frac{1}{2}, \frac{\epsilon}{2} \right\}$$

$$\text{Then } |x-2| < \delta \Rightarrow |x-2| < \frac{\epsilon}{2} \Rightarrow \left| \frac{1}{1-x} + 1 \right| < \epsilon$$

$$\Rightarrow \lim_{x \rightarrow 2} \frac{1}{1-x} = -1$$

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$$(b) \quad \lim_{x \rightarrow 1} \frac{x}{1+x} = \frac{1}{2}$$

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soln: Let $\epsilon > 0$ be given

$$\text{to find } \epsilon > 0 \text{ s.t. } \left| \frac{x}{1+x} - \frac{1}{2} \right| = \left| \frac{2x-1-x}{x+1} \right|$$

$$= \frac{|x-1|}{(x+1)} \quad \text{--- (I)}$$

$$\text{Now if } |x-1| < \frac{1}{2} \Rightarrow \frac{1}{2} < x < \frac{3}{2}$$

$$\Rightarrow \frac{3}{2} < x+1 \Rightarrow \frac{1}{x+1} < \frac{2}{3}$$

$$\left| \frac{x}{1+x} - \frac{1}{2} \right| < \frac{2}{3}|x-1| \quad \text{--- (II)}$$

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Now $\left| \frac{x}{1+x} - \frac{1}{2} \right| < \epsilon$ i.e. $\frac{2}{3} |x-1| < \epsilon \Rightarrow |x-1| < \frac{3}{2} \epsilon$

choose: $\delta = \min \left\{ \frac{1}{2}, \frac{3}{2} \epsilon \right\}$ then

$$|x-1| < \delta \Rightarrow |x-1| < \frac{1}{2}$$

$$\Rightarrow \left| \frac{x}{1+x} - \frac{1}{2} \right| < \frac{2}{3} |x-1|$$

Now $\delta = \min \left\{ \frac{1}{2}, \frac{3}{2} \epsilon \right\} \Rightarrow |x-1| < \delta$
 $\Rightarrow |x-1| < \frac{3}{2} \epsilon$

$$\Rightarrow \left| \frac{x}{1+x} - \frac{1}{2} \right| < \frac{2}{3} \cdot \frac{3}{2} \epsilon = \epsilon$$

(c) $\lim_{x \rightarrow 0} \frac{x^2}{|x|} = 0$

Let $\epsilon > 0$ be given to find $\delta > 0$ s.t.

$$|x-0| < \delta \Rightarrow \left| \frac{x^2}{|x|} - 0 \right| < \epsilon$$

Consider $\left| \frac{x^2}{|x|} - 0 \right| = \frac{x^2}{|x|} = \frac{|x|^2}{|x|} = |x|$

$\therefore (|x| \neq 0)$

Now $\left| \frac{x^2}{|x|} - 0 \right| < \epsilon$ i.e. $|x| < \epsilon$ choose $\delta = \epsilon$

then $|x-0| < \delta \Rightarrow \left| \frac{x^2}{|x|} - 0 \right| < \epsilon$

(d) $\lim_{x \rightarrow 1} \frac{x^2 - x + 1}{x+1} = \frac{1}{2}$

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Let $\epsilon > 0$ be given

Consider $\left| \frac{x^2 - x + 1}{x+1} - \frac{1}{2} \right| = \left| \frac{2x^2 - 2x + 2 - x - 1}{2(x+1)} \right|$

$$= \left| \frac{2x^2 - 3x + 1}{2(x+1)} \right| = \frac{|2x-1||x-1|}{2(x+1)} \quad \text{--- (1)}$$

Q.L. $|x-1| < \delta$ then $|2x-1| < 2\delta + 1$

$$\text{also } x > 1 \Rightarrow x+1 > 2 \Rightarrow \frac{1}{x+1} < \frac{1}{2}$$

$$\therefore \left| \frac{x^2 - x + 1}{x+1} - \frac{1}{2} \right| < \frac{5}{2} |x-1|$$

$$\text{Now } \left| \frac{x^2 - x + 1}{x+1} - \frac{1}{2} \right| < \epsilon \quad \text{if} \quad \frac{5}{2} |x-1| < \epsilon$$

$$\Rightarrow |x-1| < \frac{2}{5} \epsilon$$

Choose $\delta = \min \left\{ 2, \frac{2}{5} \epsilon \right\}$, then we can show as in above questions, that

$$\left| \frac{x^2 - x + 1}{x+1} - \frac{1}{2} \right| < \epsilon$$

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Q11 Show that following limits donot exist.

(a) $\lim_{x \rightarrow 0} \frac{1}{x^2} \quad (x > 0)$, here $f(x) = \frac{1}{x^2}$

Consider a sequence $\langle x_n \rangle = \frac{1}{n}$

$$\text{then } \lim_{n \rightarrow \infty} x_n = 0$$

$$f(x_n) = n^2 \text{ and } \lim_{n \rightarrow \infty} f(x_n) \text{ is not finite}$$

$\Rightarrow \langle f(x_n) \rangle$ is not convergent.

Hence Above limit does not exist.

(b) $\lim_{x \rightarrow 0} \frac{1}{\sqrt{x}} \quad (x > 0)$ [Hint $x_n = \frac{1}{n^2}$]

(c) $\lim_{x \rightarrow 0} (x + \text{sgn}(x))$, Here $f(x) = x + \text{sgn}(x)$

Consider sequences $\langle x_n \rangle = \frac{1}{n}$ & $\langle y_n \rangle = -\frac{1}{n}$

then $\langle x_n \rangle \rightarrow 0$ and $\langle y_n \rangle \rightarrow 0$

$$\text{and } f(x_n) = \frac{1}{n} + \text{sgn}(x_n) = \frac{1}{n} + 1$$

$$f(y_n) = -\frac{1}{n} + \text{sgn}(y_n) = -\frac{1}{n} - 1$$

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$$\langle f(x_n) \rangle \rightarrow 1 \quad \text{and} \quad \langle f(y_n) \rangle \rightarrow -1$$

but $(1 \neq -1)$

(d) $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x^2}\right)$. Here $f(x) = \sin\left(\frac{1}{x^2}\right)$

choose $x_n = \frac{1}{\sqrt{n\pi}}$ and $y_n = \frac{1}{\sqrt{2n\pi + \frac{\pi}{2}}}$

Then $\langle x_n \rangle \rightarrow 0$, $\langle y_n \rangle \rightarrow 0$

$f(x_n) = \sin n\pi = 0 \quad \forall n \in \mathbb{N}$

$f(y_n) = \sin\left(2n\pi + \frac{\pi}{2}\right) = 1 \quad \forall n \in \mathbb{N}$

$\langle f(x_n) \rangle \rightarrow 0 \quad \& \quad \langle f(y_n) \rangle \rightarrow 1 \quad \forall n \in \mathbb{N}$

limit does not exist.

Q12. soln :- Let $\epsilon > 0$ $L =$ given , to find $\delta > 0$ s.t
 $|x-0| < \delta \Rightarrow |g(x) - L| < \epsilon$

As $\lim_{x \rightarrow 0} f(x) = L$, $\exists \delta' > 0$ s.t $|x-0| < \delta'$
 $\Rightarrow |f(x) - L| < \epsilon$

Let us replace x by ax

We see $|ax-0| < \delta' \Rightarrow |f(ax) - L| < \epsilon$

$\Rightarrow |x| < \frac{\delta'}{a} \Rightarrow |g(x) - L| < \epsilon$

choose $\delta = \frac{\delta'}{a} > 0$ then

$0 < |x| < \delta \Rightarrow |g(x) - L| < \epsilon$

$\therefore \lim_{x \rightarrow 0} g(x) = L$

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Q13. Soln : (a) Let $\epsilon > 0$ be given.

$$\lim_{x \rightarrow c} (f(x))^2 = 0, \exists \delta > 0 \text{ s.t. } |x - c| < \delta$$

$$\Rightarrow |(f(x))^2 - 0| < \epsilon^2$$

$$\Rightarrow |f(x)|^2 < \epsilon^2$$

$$\Rightarrow |f(x)| < \epsilon$$

$$\lim_{x \rightarrow c} f(x) = 0$$

(b) define $f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ -1 & \text{if } x \notin \mathbb{Q} \end{cases}$

$$\text{then } (f(x))^2 = 1 \quad \forall x \in \mathbb{R} \Rightarrow \lim_{x \rightarrow c} (f(x))^2 = 1 \neq 0$$

but $\lim f(x)$ does not exist.

Q14 Let $f(x) = \begin{cases} x & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$

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(a) Show that f has a limit at $x=0$.

$$\text{Now } f(x_n) = \begin{cases} x_n & x_n \in \mathbb{Q} \\ 0 & x_n \notin \mathbb{Q} \end{cases}$$

$$\lim f(x_n) = \begin{cases} \lim x_n & x_n \in \mathbb{Q} \\ 0 & x_n \notin \mathbb{Q} \end{cases}$$

$$\Rightarrow \lim f(x_n) = \begin{cases} 0 & x_n \in \mathbb{Q} \\ 0 & x_n \notin \mathbb{Q} \end{cases} = 0$$

$$\therefore \lim_{x \rightarrow 0} f(x) = 0$$

(b) Let $c \neq 0$ Then $\exists$ sequence $\langle x_n \rangle$ of rationals and sequence $\langle y_n \rangle$ of irrationals s.t.

$$\langle x_n \rangle \rightarrow c \text{ and } \langle y_n \rangle \rightarrow c$$

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$$f(x_n) = x_n \quad \forall n \in \mathbb{N}$$

$$f(y_n) = 0 \quad \forall n \in \mathbb{N}$$

$$\lim f(x_n) = \lim x_n = c$$

$$\text{and } \lim f(y_n) = \lim 0 = 0$$

$$\text{as } c \neq 0 \quad \therefore \lim f(x_n) \neq \lim f(y_n)$$

$\therefore \lim f$ does not exist.
 $x \rightarrow c \neq 0$

Q15 (i) $f: \mathbb{R} \rightarrow \mathbb{R}$

(ii) $f_1: I \rightarrow \mathbb{R}$, $f_1(x) = f(x) \quad \forall x \in I$

(iii) I is open set and $c \in I$

show that $\lim_{x \rightarrow c} f = L \iff \lim_{x \rightarrow c} f_1 = L$

Soln: Assume $\lim_{x \rightarrow c} f = L$

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Let $\epsilon > 0$ be given

$$\lim_{x \rightarrow c} f = L \quad \therefore \exists \delta_1 > 0 \text{ s.t. } |x - c| < \delta_1 \Rightarrow |f(x) - L| < \epsilon \quad \text{--- (1)}$$

Now $c \in I$ and $c \in V_{\delta_1}(c) \Rightarrow c \in I \cap V_{\delta_1}(c)$

As I is an open set and $V_{\delta_1}(c)$ is also open set

$\therefore I \cap V_{\delta_1}(c)$ is an open set and $c \in I \cap V_{\delta_1}(c)$

$\therefore$ by def $\exists \delta > 0$ s.t. $V_\delta(c) \subseteq I \cap V_{\delta_1}(c)$

$$\Rightarrow V_\delta(c) \subseteq I \text{ and } V_\delta(c) \subseteq V_{\delta_1}(c)$$

Now $x \in V_\delta(c) \Rightarrow x \in I$ and $x \in V_{\delta_1}(c)$

Now $f_1(x) = f(x) \quad \forall x \in I$

$$\Rightarrow |x - c| < \delta \Rightarrow |f_1(x) - L| < \epsilon \quad \forall x \in I$$

$$\therefore \lim_{x \rightarrow c} f_1 = L$$

Let us now Assume that $\lim_{x \rightarrow c} f_1 = L$

Let $\epsilon > 0$ be given, to find a $\delta > 0$ s.t.

$$|x - c| < \delta \Rightarrow |f(x) - L| < \epsilon$$

As $\lim_{x \rightarrow c} f_1 = L \therefore \forall \epsilon > 0, \exists \delta > 0$ s.t.

$$|x - c| < \delta \Rightarrow |f_1(x) - L| < \epsilon, \quad x \in I$$

$$\Rightarrow |f(x) - L| < \epsilon \quad \left(\begin{array}{l} f_1(x) = f(x) \\ \forall x \in I \end{array} \right)$$

$$\Rightarrow |x - c| < \delta \Rightarrow |f(x) - L| < \epsilon$$

$$\lim_{x \rightarrow c} f = L$$

Q16 (i) $f: \mathbb{R} \rightarrow \mathbb{R}$

(ii) $f_2: J \rightarrow \mathbb{R}, f_2(x) = f(x) \quad \forall x \in J$

(iii) J is closed interval and $c \in J$

Soln: Assume $\lim_{x \rightarrow c} f = L$

To show $\lim_{x \rightarrow c} f_2 = L$

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As J is a closed Interval $\therefore J \in [a, b]$

case (i). $c = a$

As $\lim_{x \rightarrow c} f = L \therefore \forall$ given any $\epsilon > 0$ we get

$$\delta > 0 \text{ s.t. that } |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$$

$$\Rightarrow a - \delta < x < a + \delta \Rightarrow |f(x) - L| < \epsilon \quad \text{--- (I)}$$

$$\Rightarrow a < x < a + \delta \Rightarrow |f(x) - L| < \epsilon, \quad x \in J$$

$$\Rightarrow a < x < a + \delta \Rightarrow |f_2(x) - L| < \epsilon$$

$$\left(f_2(x) = f(x) \quad \forall x \in J \right)$$

$$\Rightarrow \lim_{x \rightarrow a} f_2 = L$$

$$\text{i.e. } \lim_{x \rightarrow c} f_2 = L$$

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P.14

SECTION 4.2

Def: BOUNDED

Let $A \subseteq \mathbb{R}$, c be a cluster point of A
 f is bounded on nbd of c if there exist
 $[c-\delta, c+\delta[$ and constant $M > 0$ s.t
 $|f(x)| \leq M \quad \forall x \in A \cap V_\delta(c)$

Theorem 1: If $A \subseteq \mathbb{R}$ and $f: A \rightarrow \mathbb{R}$ has a limit at c
then f is bounded on some nbd of c

Proof: If $\lim_{x \rightarrow c} f = L$ then for $\epsilon = 1 > 0$, $\exists \delta > 0$ s.t
 $|x - c| < \delta \Rightarrow |f(x) - L| < 1 \quad \text{--- (1)}$

Consider $|f(x)| = |f(x) - L + L|$
 $\leq |f(x) - L| + |L|$

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$= |L| + 1 \quad \text{(from (1))}$

$|f(x)| \leq |L| + 1 \quad \forall x \in A \cap V_\delta(c)$

Then $|f(x)| \leq M \quad \forall x \in A \cap V_\delta(c) \quad [|L| + 1 = M > 0]$

Theorem 2: ALGEBRA OF LIMITS

Let $A \subseteq \mathbb{R}$, c be a cluster pt. of A

Let $f + g: A \rightarrow \mathbb{R}$, and l and $m \in \mathbb{R}$

Let $\lim_{x \rightarrow c} f(x) = l$ & $\lim_{x \rightarrow c} g(x) = m$ Then

(a) Sum Thm: $\lim_{x \rightarrow c} (f+g)(x) = l+m$

(b) product Thm: $\lim_{x \rightarrow c} (fg)(x) = lm$

Proof (a) To show $\lim_{x \rightarrow c} (f+g)(x) = l+m$

Let $\langle x_n \rangle$ be a sequence in A s.t. $\langle x_n \rangle \rightarrow c$

$$\left. \begin{array}{l} \text{As } \lim_{x \rightarrow c} f(x) = l \quad \therefore \langle f(x_n) \rangle \rightarrow l \\ \text{As } \lim_{x \rightarrow c} g(x) = m \quad \therefore \langle g(x_n) \rangle \rightarrow m \end{array} \right\} \begin{array}{l} \text{by Seq.} \\ \text{CRITERION Thm} \\ \text{of Limits} \end{array}$$

$$\therefore \langle f(x_n) + g(x_n) \rangle \rightarrow l+m$$

(by Algebra of convergence)

$$\text{i.e. } \langle (f+g)(x_n) \rangle \rightarrow l+m$$

$$\therefore \lim_{x \rightarrow c} (f+g)(x) = l+m$$

(again by Seqⁿ criterion of limits)

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(b) do yourself

Assignment : State & prove the following using sequential criterion of limits

1) Difference Thm 2) Scalar Mult thm 3) Quotient Thm.

Cor of Product Thm :

$$\text{SQUARE THM : } \lim_{x \rightarrow c} f^2(x) = l^2$$

Reason take $g = f$ & $m = l$ in product Thm

$$\text{Restatement : } \lim_{x \rightarrow c} f^2(x) = \left\{ \lim_{x \rightarrow c} f(x) \right\}^2$$

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P.16

MODULUS THM Let $A \subseteq \mathbb{R}$, c be a cluster pt. of A
Let $f: A \rightarrow \mathbb{R}$ & let $l \in \mathbb{R}$
If $\lim_{x \rightarrow c} f(x) = l$, then $\lim_{x \rightarrow c} |f(x)| = |l|$

Proof: Let $\epsilon > 0$ be given

$$[\text{T.S. } \exists \delta > 0 \text{ s.t. } x \in A, |x - c| < \delta \Rightarrow |f(x) - l| < \epsilon]$$

$$\text{consider } ||f(x)| - |l|| = ||f(x)| - |l|| \\ \leq |f(x) - l| \quad - (1)$$

$$\text{As } \lim_{x \rightarrow c} f(x) = l \quad \therefore \text{cor to } \epsilon > 0, \exists \delta > 0 \text{ s.t. } x \in A$$

$$|x - c| < \delta \Rightarrow |f(x) - l| < \epsilon \quad - (2)$$

$$(1) \text{ \& } (2) \Rightarrow x \in A, |x - c| < \delta \Rightarrow ||f(x)| - |l|| < \epsilon$$

As $\epsilon > 0$ is arbitrary

$$\therefore \lim_{x \rightarrow c} |f(x)| = \left| \lim_{x \rightarrow c} f(x) \right| = |l|$$

SQUARE ROOT THM : Let $A \subseteq \mathbb{R}$ & c be a cluster pt. of A . Let $f: A \rightarrow \mathbb{R}$ & let $l \in \mathbb{R}$ & let $f(x) \geq 0$
 $\forall x \in A$. If $\lim_{x \rightarrow c} f(x) = l$, then $\lim_{x \rightarrow c} (\sqrt{f})(x) = \sqrt{l}$

Proof: As $f(x) \geq 0 \quad \forall x \in A \quad \therefore \lim_{x \rightarrow c} f(x) = l \geq 0$

Case (i) $l > 0$. Let $\epsilon > 0$ be given

$$[\text{T.S. } \exists \delta > 0 \text{ s.t. } x \in A, |x - c| < \delta \Rightarrow |(\sqrt{f})(x) - \sqrt{l}| < \epsilon]$$

$$\text{Consider } |(\sqrt{f})(x) - \sqrt{l}| = \left| \frac{f(x) - l}{(\sqrt{f})(x) + \sqrt{l}} \right| \\ = |f(x) - l| \times \frac{1}{|(\sqrt{f})(x) + \sqrt{l}|}$$

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R17

$$\lim_{x \rightarrow c} f(x) = l \quad \text{iff} \quad \text{for each } \epsilon > 0, \exists \delta > 0 \text{ s.t.} \\ |x - c| < \delta \Rightarrow |f(x) - l| < \epsilon \quad \text{--- (2)}$$

∴ eq (1) gives

$$|\sqrt{f(x)} - \sqrt{l}| \leq |f(x) - l| \cdot \frac{1}{\sqrt{l}} \\ < \epsilon \cdot \sqrt{l} \cdot \frac{1}{\sqrt{l}} \quad \text{--- from (2)} \\ = \epsilon$$

$$\therefore |x - c| < \delta \Rightarrow |\sqrt{f(x)} - \sqrt{l}| < \epsilon$$

$$\therefore \lim_{x \rightarrow c} \sqrt{f(x)} = \sqrt{l}$$

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Case (ii) $l = 0$

$$\lim_{x \rightarrow c} \sqrt{f(x)} = 0 \quad \text{iff} \quad \text{for each } \epsilon^2 > 0, \exists \delta > 0$$

$$\text{s.t. } |x - c| < \delta \Rightarrow |f(x) - 0| < \epsilon^2$$

$$\Rightarrow |x - c| < \delta \Rightarrow |\sqrt{f(x)} - 0| < \epsilon$$

$$\Rightarrow \lim_{x \rightarrow c} \sqrt{f(x)} = \sqrt{0} = 0 = \sqrt{\lim_{x \rightarrow c} f}$$

Theorem: Let $A \subseteq \mathbb{R}$, Let $f: A \rightarrow \mathbb{R}$, Let $c \in \mathbb{R}$

be a cluster pt. of A . If $a \leq f(x) \leq b \quad \forall x \in A$,

$x \neq c$ and if $\lim_{x \rightarrow c} f$ exists, then $a \leq \lim_{x \rightarrow c} f \leq b$.

Proof: Let $\lim_{x \rightarrow c} f(x) = L$

Let $\langle x_n \rangle \rightarrow c \Rightarrow \lim_{n \rightarrow \infty} f(x_n) = L$

Then $a \leq f(x_n) \leq b \quad \forall n \in \mathbb{N}$

$\Rightarrow a \leq \lim_{n \rightarrow \infty} f(x_n) \leq b \quad \forall n \in \mathbb{N}$

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P.18

SQUEEZE THEOREM

Let $A \subseteq \mathbb{R}$, Let $f, g, h : A \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A . If $f(x) \leq g(x) \leq h(x) \quad \forall x \in A, x \neq c$ and if $\lim_{x \rightarrow c} f = L = \lim_{x \rightarrow c} h$ then $\lim_{x \rightarrow c} g = L$.

Proof: Let $\epsilon > 0$ be given

$$[\text{I.e. } \exists \delta > 0 \text{ s.t. } |x - c| < \delta \Rightarrow L - \epsilon < g(x) < L + \epsilon]$$

As $\lim_{x \rightarrow c} f(x) = L$ iff for each $\epsilon > 0$, $\exists \delta_1 > 0$ s.t.
 $|x - c| < \delta_1 \Rightarrow L - \epsilon < f(x) < L + \epsilon$

+ $\lim_{x \rightarrow c} h(x) = L$ iff for each $\epsilon > 0$, $\exists \delta_2 > 0$ s.t.
 $|x - c| < \delta_2 \Rightarrow L - \epsilon < h(x) < L + \epsilon$

choose $\delta = \min(\delta_1, \delta_2)$

then $|x - c| < \delta \Rightarrow L - \epsilon < f(x) < L + \epsilon$
 $\Rightarrow L - \epsilon < h(x) < L + \epsilon$

Now $f(x) \leq g(x) \leq h(x) \quad \forall x \in V_\delta(c)$

$\Rightarrow L - \epsilon < g(x) < L + \epsilon \quad \forall x \in V_\delta(c)$

$$\lim_{x \rightarrow c} g(x) = L$$

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Theorem 4.2.9 Proof:-

Let $\lim_{x \rightarrow c} f = L$. then $L > 0 \Rightarrow \frac{L}{2} > 0$

As $\lim_{x \rightarrow c} f = L \quad \therefore \exists \delta > 0$ s.t. $|x - c| < \delta \Rightarrow |f(x) - L| < \frac{L}{2}$

$\forall x \in A \cap V_\delta(c), x \neq c$, then $\frac{L}{2} < f(x) < \frac{3L}{2}$

$\Rightarrow 0 < \frac{L}{2} < f(x) \quad \forall x \in A \cap V_\delta(c), x \neq c$

Similarly we can show $L < 0$.

PROBLEM SET - 2

Q4 soln: Pg 110 Let $x_n = \frac{1}{2n\pi}$, $y_n = \frac{1}{2n\pi + \frac{\pi}{2}}$

Then $\langle x_n \rangle \rightarrow 0$, $\langle y_n \rangle \rightarrow 0$

Here $f(x_n) = \cos(2n\pi) = 1 \quad \forall n \in \mathbb{N}$

$f(y_n) = \cos(2n\pi + \frac{\pi}{2}) = 0 \quad \forall n \in \mathbb{N}$

$\lim_{n \rightarrow \infty} f(x_n) = 1 \neq 0 = \lim_{n \rightarrow \infty} f(y_n)$

$\therefore \lim_{x \rightarrow 0} \cos(\frac{1}{x})$ does not exist

Now $\lim_{x \rightarrow 0} x \cos(\frac{1}{x})$; Let $\epsilon > 0$ be given

consider $|x \cos \frac{1}{x} - 0| \leq |x - 0|$

choose $\delta = \epsilon$ then $|x - 0| < \delta \Rightarrow |x \cos(\frac{1}{x}) - 0| < \epsilon$

$\therefore \lim_{x \rightarrow 0} x \cos \frac{1}{x} = 0$

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on A.S.R. and

Q5 soln: Pg 110 Let f, g be defined on $A \subseteq \mathbb{R}$ and

Let c be a cluster point

let $\epsilon > 0$ be given

[T.S. $\exists \delta > 0$ s.t. $|x - c| < \delta \Rightarrow |(fg)(x) - 0| < \epsilon$]

consider $|f(x) \cdot g(x) - 0| = |f(x)| |g(x)| \quad \text{--- (1)}$

As f is bounded on nbd. of c

$\exists M > 0$ and $\delta > 0$ s.t. $|f(x)| \leq M \quad \forall x \in A \cap V_\delta(c)$

d. $\lim_{x \rightarrow c} g(x) = 0$, $\forall \frac{\epsilon}{M} > 0$, $\exists \delta > 0$ s.t.

$|x - c| < \delta \Rightarrow |g(x)| < \frac{\epsilon}{M}$

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P.20

Now choose $\delta = \min(\delta_1, \delta_2)$

$$\left. \begin{aligned} \text{Then } |x-c| < \delta &\Rightarrow |f(x)| \leq M \\ &\Rightarrow |g(x)| \leq \frac{\epsilon}{M} \end{aligned} \right\} \text{--- (2)}$$

$$\therefore \text{eqn (1) gives } |f(x)g(x)| = |f(x)||g(x)| < M \cdot \frac{\epsilon}{M} = \epsilon$$

$$\therefore |f(x)g(x)| < \epsilon \quad \forall x \in A \cap V_\delta(c)$$

$$\lim_{x \rightarrow c} (fg)(x) = 0$$

Q9 Soln (a) Given $\lim_{x \rightarrow c} f$ and $\lim_{x \rightarrow c} (f+g)$ exists
Pg III

To show that $\lim_{x \rightarrow c} g$ exists

$$\text{Let } g = (f+g) - f$$

$$\Rightarrow \lim_{x \rightarrow c} g = \lim_{x \rightarrow c} (f+g) - \lim_{x \rightarrow c} f$$

As $\lim_{x \rightarrow c} (f+g)$ & $\lim_{x \rightarrow c} f$ exist

$$\Rightarrow \lim_{x \rightarrow c} g \text{ exists.}$$

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Soln (b) If $\lim_{x \rightarrow c} f$ and $\lim_{x \rightarrow c} fg$ exists, does it

follow that $\lim_{x \rightarrow c} g$ exist?

$$\text{Let } f(x) = x \quad \forall x \in \mathbb{R}$$

$$g(x) = \sin \frac{1}{x} \quad x \neq 0$$

Then $\lim_{x \rightarrow 0} f(x)$ exist but $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ does not exist

also $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$ exist.

$\therefore \lim_{x \rightarrow c} f$ exist and $\lim_{x \rightarrow c} fg$ exist but need not

Q11 (c) Soln: Let $\langle x_n \rangle$ be the sequence defined as

Pg 111

$$x_n = \frac{2}{n\pi} \quad \text{then} \quad \lim_{n \rightarrow \infty} x_n = 0$$

$$\sin\left(\frac{1}{x_n}\right) = \sin\left(\frac{n\pi}{2}\right) = \begin{cases} 1 & \text{when } n = 1, 5, 9, \dots \\ 0 & n = \text{even} \\ -1 & \text{when } n = 3, 7, 11, \dots \end{cases}$$

$$\therefore \operatorname{Sgn} \sin\left(\frac{1}{x_n}\right) = \begin{cases} 1 & n = 1, 5, 9, \dots \\ 0 & n = \text{even} \\ -1 & n = 3, 7, 11, \dots \end{cases}$$

Then clearly $\operatorname{Sgn} \sin\left(\frac{1}{x_n}\right)$ does not exist.

$\therefore \lim_{x \rightarrow 0} \operatorname{Sgn} \sin\left(\frac{1}{x_n}\right)$ does not exist.

Q12 (a) Let $f: \mathbb{R} \rightarrow \mathbb{R}$, s.t. that

$$f(x) + f(y) = f(x+y) \quad \forall x, y \in \mathbb{R}$$

Assume that $\lim_{x \rightarrow 0} f = L$ (exists) T. show $L = 0$

$$\text{Now } f(2x) = f(x) + f(x) \\ = 2f(x)$$

$$\lim_{x \rightarrow 0} f(2x) = \lim_{x \rightarrow 0} 2f(x) \\ = 2 \lim_{x \rightarrow 0} f(x)$$

$$L = 2L \Rightarrow 2L - L = 0 \Rightarrow L = 0$$

(b) Consider $f(x) = f(x-c) + f(c)$

$$\Rightarrow \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} f(x-c) + f(c)$$

$$= L + f(c)$$

(which is finite & exists)

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